Machine Learning and Big Data - DATA622

Foundation models

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Generative Adversarial Networks (GANs)

Identifying the data manifold

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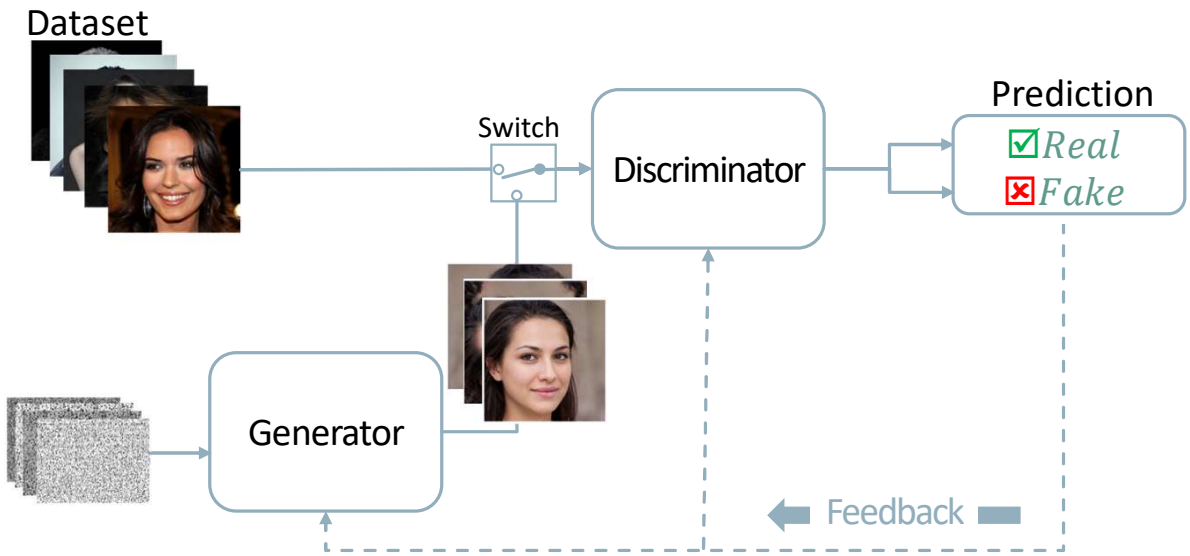
Which photo is fake?

Both are fake

Source: TLL

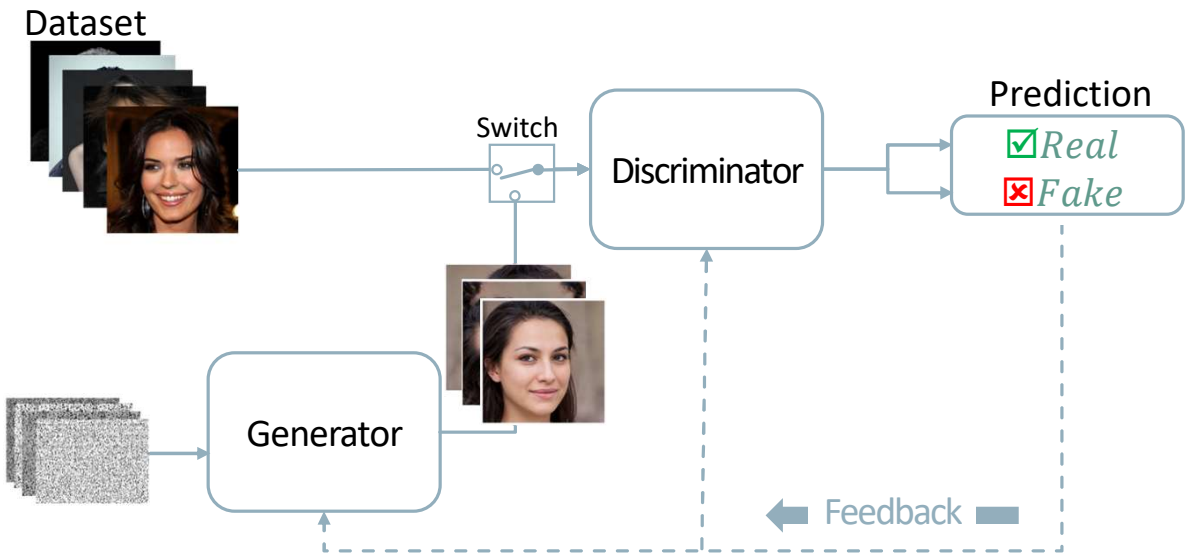
- Source: NVIDIA. A Style-Based Generator Architecture for Generative Adversarial Networks https://arxiv.org/abs/1812.04948

GAN Architecture



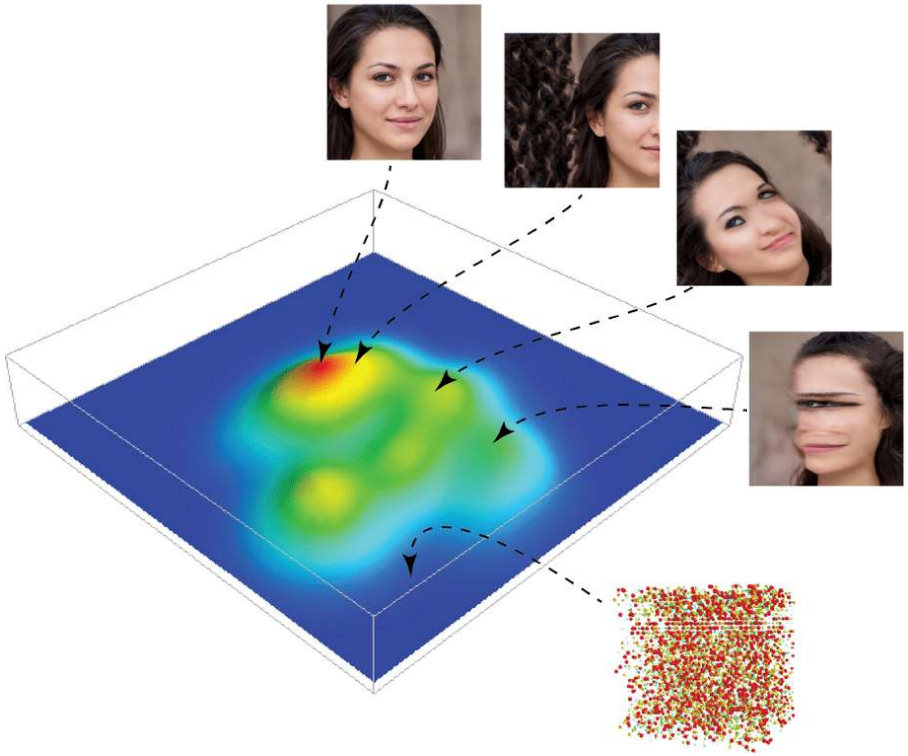
Source: TLL

GAN Concept


Dataset
Switch
Discriminator
Prediction
Real
Fake
Generator
Feedback

Generative Adversarial Networks

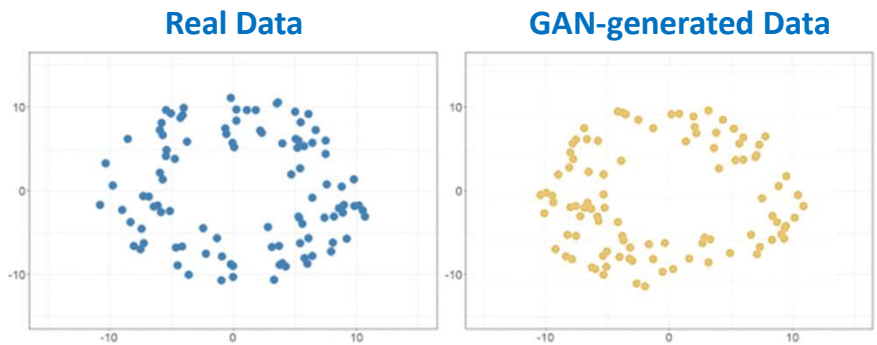
- Generative Adversarial Networks (GAN) are a neural networks architecture which contains two neural networks: a generator and a discriminator.
- During training the GAN will map the layout of the manifold that defines the training data
- After training, the generator will be able to generate data similar to the training data set



Source: Images from Network Bending: Manipulating The Inner Representations of Deep Generative Models, Terence Broad, et al.

Generate Synthetic data using a GAN

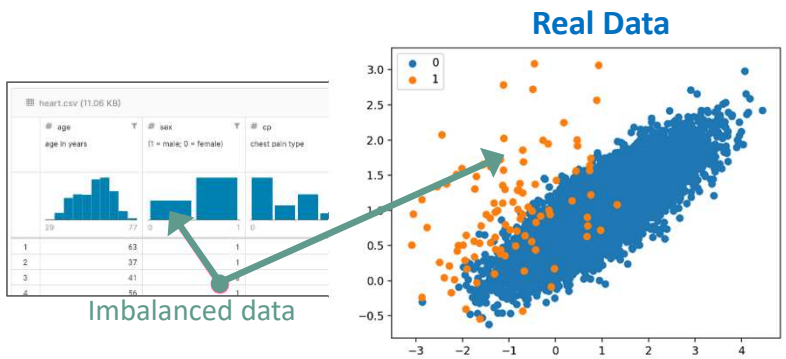
- Synthetic data can be generated from the original training data using a GAN (or similar methods)
- The synthetic data retains the “shape” (the structure) of the original data but is not the same data
- Synthetic data allows non-prod data to be used for model training
- Synthetic data offers the potential for privacy-preserving data

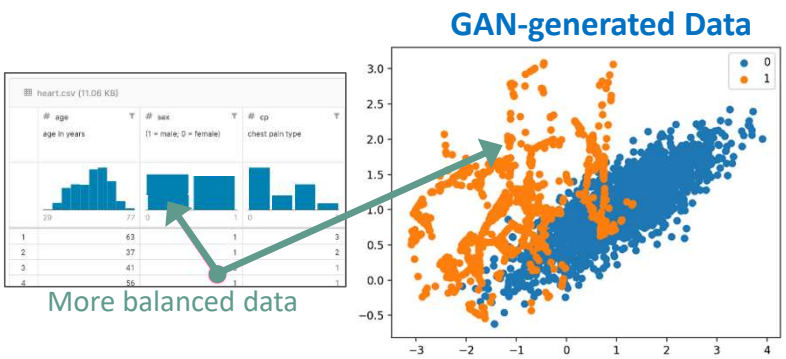


Source: Synthetic Data – A Privacy Mirage, Theresa Stadler, et al. Dec. 2020

Reducing Imbalanced Data

- One cause of bias (there are many) is imbalanced data sets.
- The challenge of working with imbalanced datasets is that the training effort is spent mostly on the dominant data class
- Training will ignore, and in turn have poor performance on, the minority class
- Typically it is performance on the minority class that is most important e.g. fraud detection
- GANs (and similar generative approaches) can generate synthetic data to balance classes





Source: SMOTE for Imbalanced Classification with Python, Machine Learning Mastery

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Large Language Models

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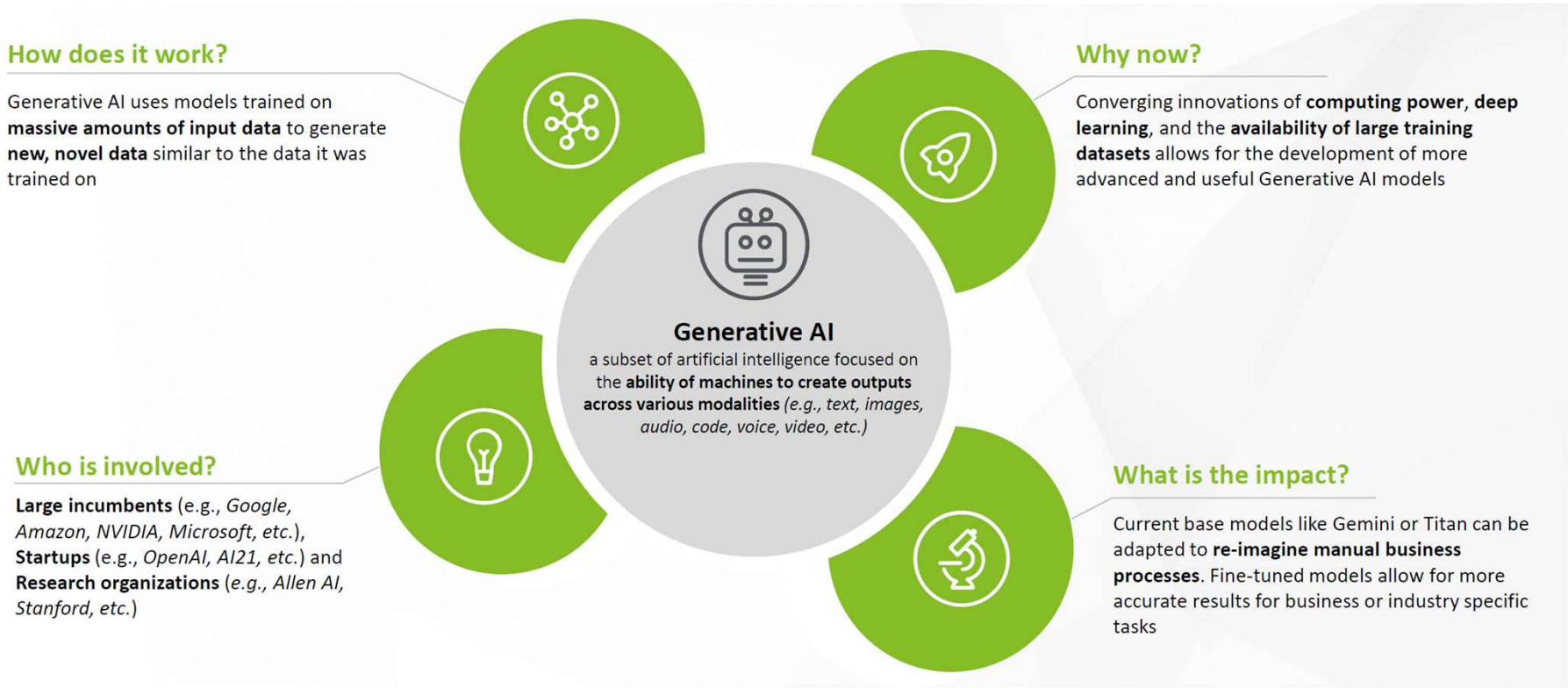
Large Language Models



Source: McKinsey

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Large Language Models

Large Language Models (LLMs)

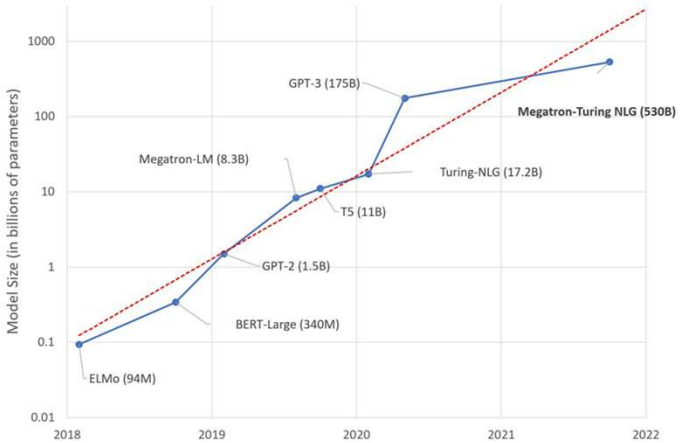
- Pre-trained with extremely large datasets – architected to scale
- Can be adopted to a wide range of downstream tasks
- LLMs are Foundation Models.

Very Large training datasets

GPT-3 Datasets Summary.

	Wikipedia	Books	Journals	Reddit links	CC	Other	Total
GB	*11.4*	*21*	*101*	*50*	**570**		753GB
Tokens	**3**	**12**	**55**	**19**	**410**		499BTokens

LLM Scaling: a new Moore's Law



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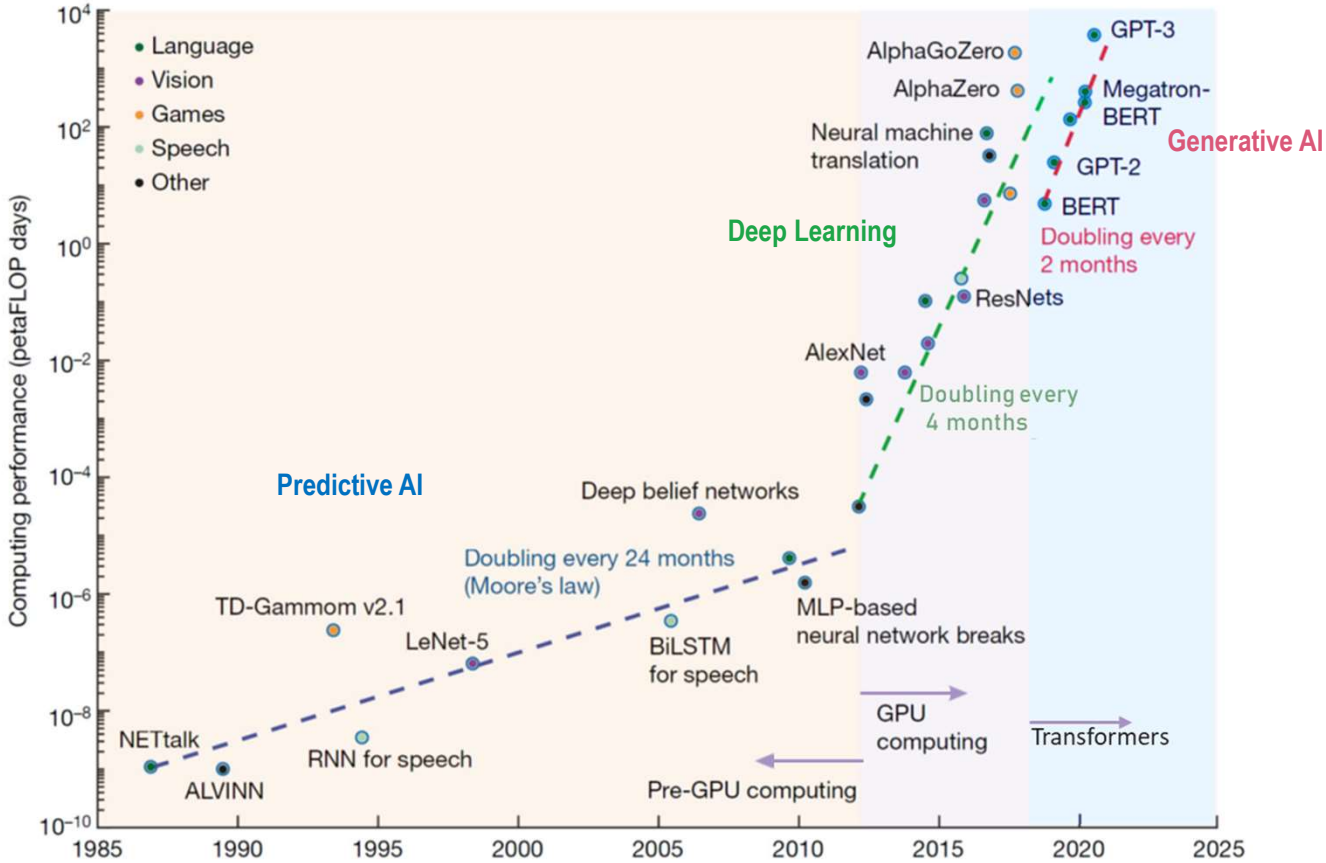
Accelerating Growth in Performance



Source: Large Language Model and the Turing Test, Terrence Sejnkowski

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Prompt Engineering

- Prompts offer a natural language approach to communicate with a model
- Provide examples for complex tasks (few-shot learning)
- Ambiguous inputs result in ambiguous outputs (GIGO)
- Define a persona (e.g., “You are a compliance expert…”)
- Be consistent (use prompt templates for similar tasks)

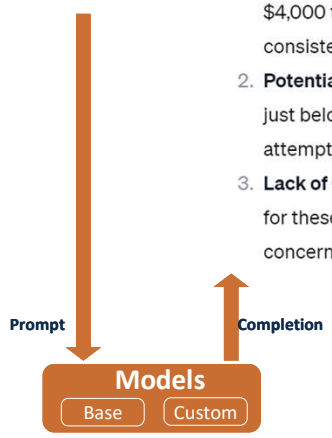
You

A customer has been a client for 5 years. They are in good standing with the financial institution and have had no suspicious activity in the past. They make relatively routine transactions that have not raised any flags. Seemingly out of nowhere, the customer makes a handful of transactions of $4,000 each. In total, they make 5 transactions two weeks apart, amounting to $20,000. What potential SAR red flags does this raise? The output should be a recommendation, followed by a numbered list, with no more than 3 items

ChatGPT

Recommendation: Given the sudden change in behavior, it is advisable to conduct a further review of the customer's transactions and consider filing a Suspicious Activity Report (SAR) if warranted. Key red flags to assess include:

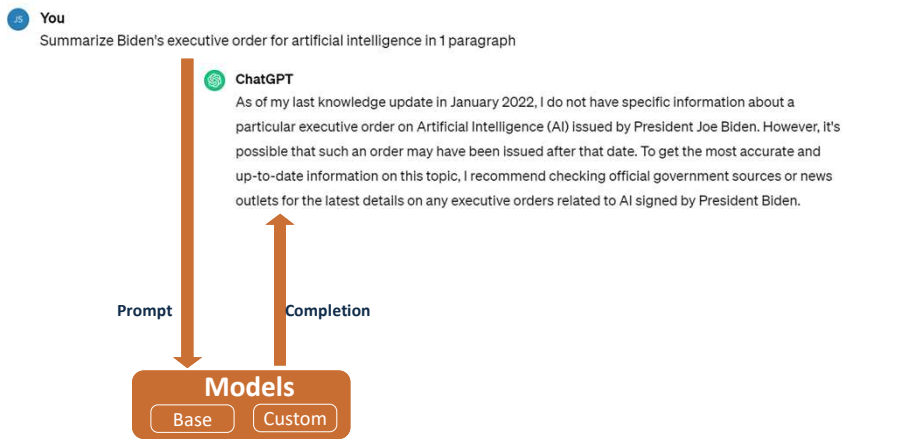
1. **Unusual Transaction Pattern**: The abrupt shift in transaction behavior from routine to a series of $4,000 transactions, occurring at regular intervals, should be carefully examined as it is not consistent with the customer's historical activity.
2. **Potential Structuring or Smurfing**: The repetitive nature of these $4,000 transactions, which are just below the $5,000 threshold for Currency Transaction Reports (CTRs), may indicate an attempt to avoid reporting requirements, potentially involving structuring or smurfing.
3. **Lack of Clear Explanation**: If the customer cannot provide a legitimate reason or documentation for these transactions, and there is no apparent business or personal need for them, it raises concerns about the source of funds and the purpose of the transactions.



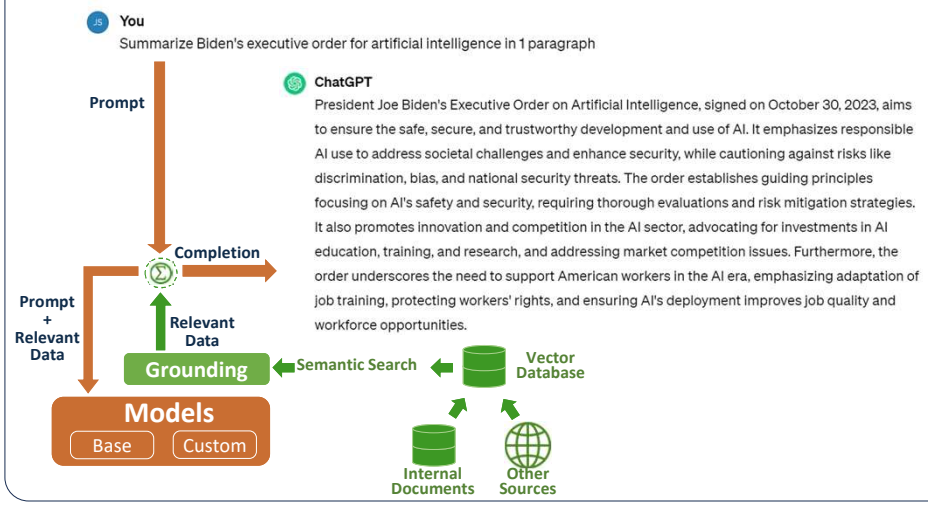
Grounding

- LLMs have a lot of knowledge but its stale, and doesn't have access to your data
- Grounding ensures the model has access to current and relevant data
- Reduces the possibility that a LLM will hallucinate
- Only relevant data is sent (minimizing potential to leak sensitive data)
- Data can be updated, without having to re-train the model
- Also called Retrieval Augmented Generation (RAG)

Prompt only (no Grounding)

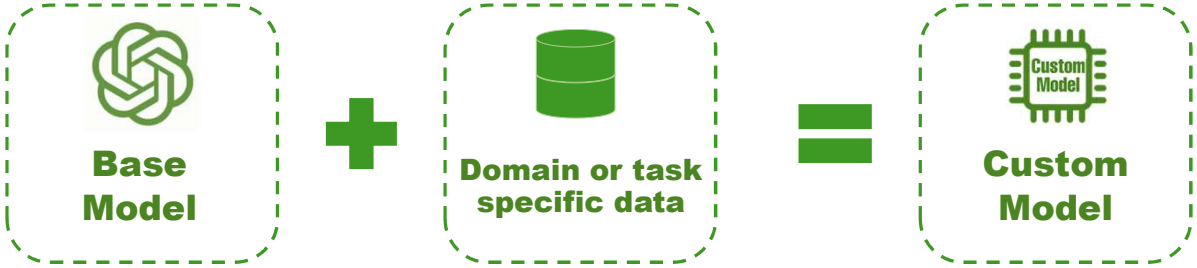


With Grounding



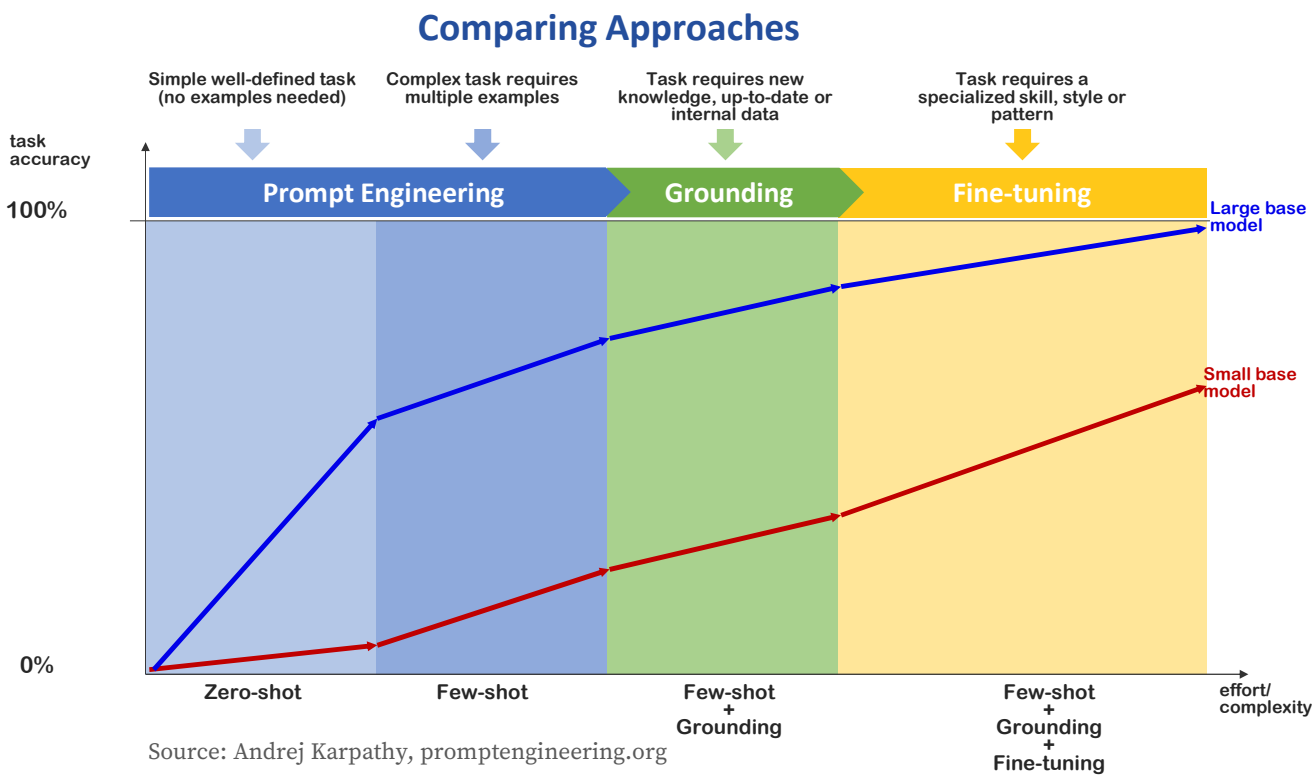
Fine-tuning

- Enhance the performance of an LLM for a specific task
- Faster and cheaper than training an LLM from scratch
- Doesn’t impact model performance for other LLM tasks
- You can add more data to the model than what can fit into a prompt (context length)
- Model learns the data, instead of just accessing it (like Grounding)



Prompt Engineering vs Grounding vs Fine-tuning

- Its not either/or: the combination of all three will optimize performance (but consider cost & complexity)
- A fine-tuned small model may perform better than a large base model for a specific skill



Source: Andrej Karpathy, promptengineering.org

Challenges of ML Today

Data
- Task-specific data
- Feature Engineering
- 'Hand-crafted'

Models
- Domain-specific
- Limited re-use
- 'Hand-crafted'

Talent
- Data Scientists
- ML Researchers
- Specialized skills

Traditional Machine Learning

Time
- Long ML lifecycles
- Hard to do quick tests
- Slow retraining cycle

Costs
- High retraining costs
- Limited enterprise scaling
- Hard to prototype until later in the project

Benefits of LLMs

Increased Velocity
- Focus shifts from training models from scratch, to fine-tune models
- Faster time to market

Increased Opportunity for AI/ML involvement
- Potential to scale to a wider pool of users to perform AI/ML
- Simple text interface and natural language instruction

Cost effective
- Scale to multiple use-cases per LLM
- Faster / lower-cost prototyping

Tapping into state-of-the-art AI
- Few-shot learners (and 'Surprisingly good without fine tuning')
- Perform tasks not explicitly trained on

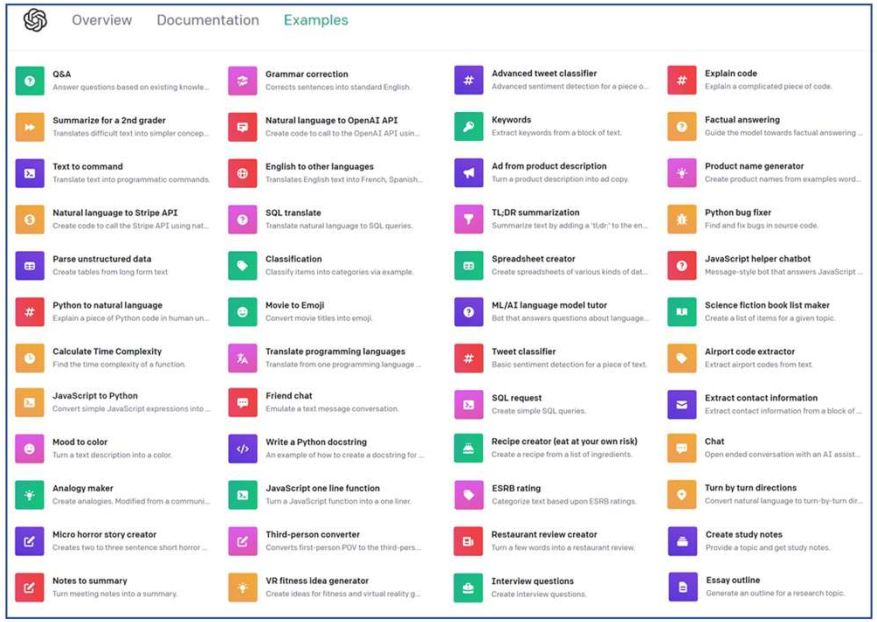
Emergent Capabilities
- Emergent capabilities that surface with LLM size
  i.e. capabilities not present in smaller models but emerge in larger models
- LLM scale highly correlated with downstream performance[1]

Benefits of LLMs

Fine Tuning
- Load foundational model
- Add task-specific prompts
- Minimal data, compute, time
- SOTA results

Embeddings
- Encode Content into dense vector for downstream use
- Use in downstream models or similarity search

In-context Learning
- "Ask" the model to perform a task as part of the input
- Provide examples to help



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Risks of LLMs

Bias Propagation
- Potential bias / toxic output
- Responsible AI is critical

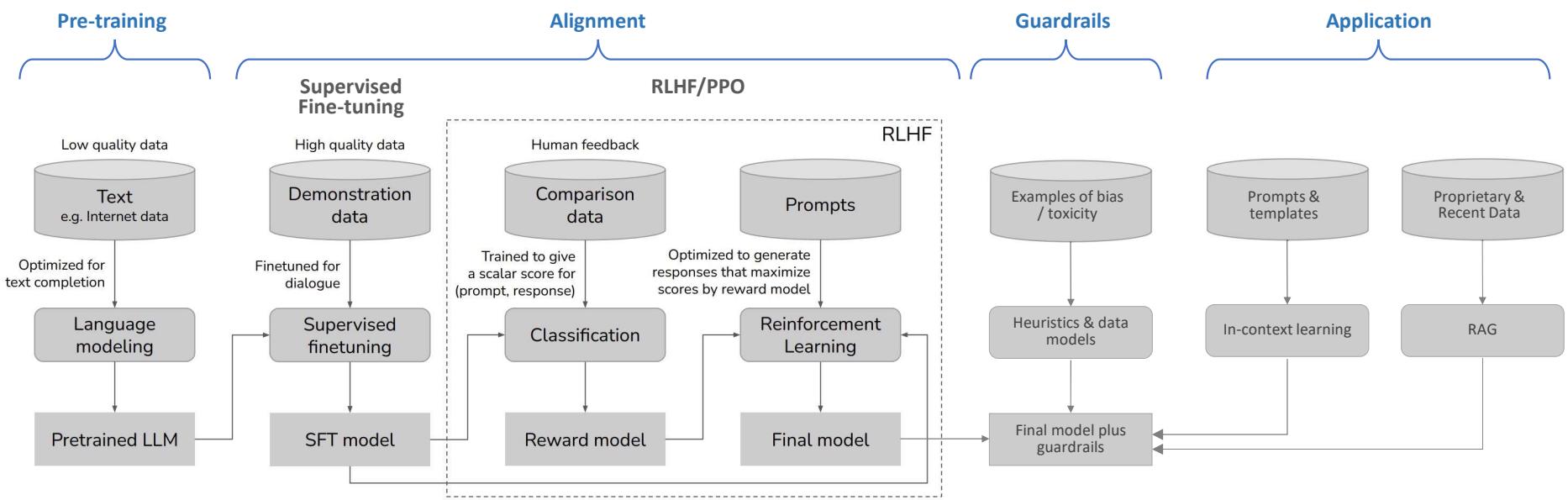
LLMs fail in subtle ways
- Hallucinations
- Evaluation / safeguards required

Increasing model scale
- Exponential growth in size
- Complexity in training / deployment

LLM costs
- Cost / Latency trade-off
- Understand & manage costs

Closed-Source models
- Legal restrictions to some models

Process of training LLMs



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Transformers

Technology underlying LLMs

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Attention

Attention Is All You Need

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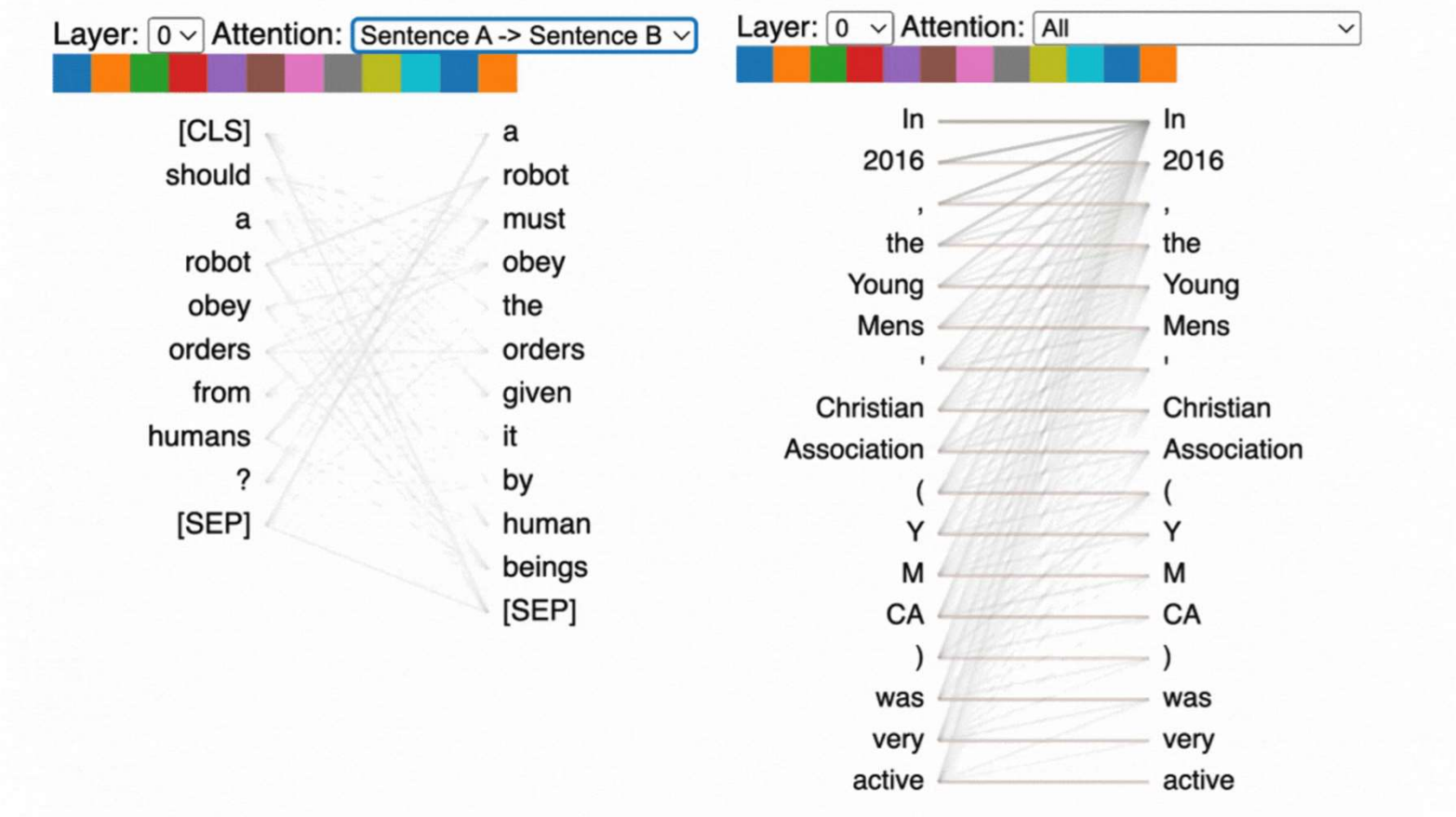
Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

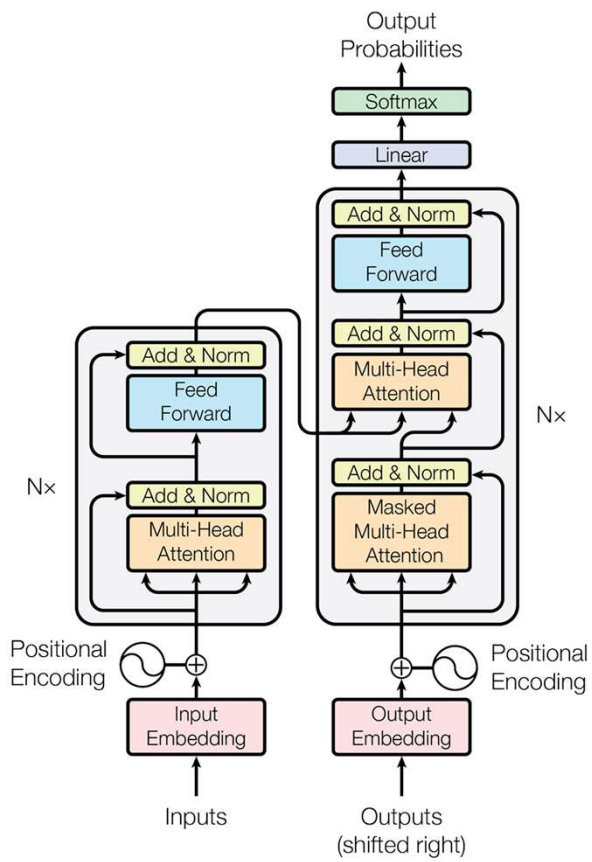
https://arxiv.org/abs/1706.03762

Attention



https://colab.research.google.com/github/tensorflow/tensor2tensor/blob/master/tensor2tensor/notebooks/hello_t2t.ipynb#scrollTo=OJKU36QAfqOC

Transformers



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